Module 7

Capital Budgeting

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

Goals

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- But to use NPV, we need a project's cash flows
- Here, combine what we know about accounting and cash flow effects (**Module 2**) with the NPV rule (**Module 6**), to decide whether or not to accept a project
- **Capital budgeting**, the process of analyzing capital investment opportunities ("projects") and deciding which ones to accept (textbook chapter 9)

Module 7, Part 1

The Three Sources of Project Cash Flow

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$$\begin{aligned}\text{Total cash flow from a project} &= \text{Cash flow from capital investments} \\ &+ \text{Operating cash flow before accounting for } \Delta\text{NWC} \\ &+ \text{Cash flow from changes in working capital}\end{aligned}$$

- Capital investments happen mostly at $t = 0$

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- Usually capitalized and depreciated over time from an accounting perspective, in this course always "straight-line" depreciation,

$$\text{Annual depreciation} = \frac{\text{Investment} - \text{Salvage value}}{\text{Expected life}}$$

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– Cash expenses for the project
– Depreciation for project-specific capital investment
Project-specific EBIT
– Taxes (tax rate times project-specific EBIT)
Project-specific net income

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- Note, revenue minus cash expenses is called the "gross profit"

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3. **Add back the depreciation tax shield**,

   $$(\text{Revenue} - \text{Cash expenses}) \times (1 - \text{tax rate}) + \text{Depreciation} \times \text{tax rate}$$

Example, Depreciation and Tax Shield in Operating Cash Flow

- A project requires an investment of \$2,000, producing gross profits (revenues less cash expenses) of \$1,100 in year 1 (and \$1,300 in year 2). The initial investment can be depreciated at \$1,000 per year. The corporate tax rate is 21%

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- Calculate the operating cash flow before accounting for ΔNWC for year 1, using the three methods,
  1. Cash in, cash out, Revenue $-$ Cash expenses $-$ Taxes
  2. Adjusted accounting profits, Net income $+$ Depreciation
  3. Add back the depreciation tax shield,
     (Revenue $-$ Cash expenses)$(1-t)$ + Depreciation $\times\, t$

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Method 1, cash in, cash out

Item	$
Gross profit	1,100
Taxes	21
Cash flow	1,079

- Method 1, the challenge is to calculate the tax correctly, $(1{,}100 - 1{,}000) \times 21\% = 21$ (gross profit is revenue minus cash expenses)

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Method 2, adjusted profits

Item	$
Gross profit	1,100
Depreciation	1,000
EBIT	100
Taxes	21
Net income	79
+ Depreciation	1,000
Cash flow	1,079

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- All three methods give \$1,079

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- Over the project's life, NWC can keep going up (or start to go down), giving additional negative (or positive) cash flows in the years beyond $t = 0$
- Working capital is often recovered at the end of the project (inventories consumed, customers pay their bills)

Example, Putting It All Together

- A company is considering buying a new copying machine which costs \$100,000 and will last 5 years. It will be depreciated straight-line over 5 years with a salvage value of \$20,000. The machine can also be sold at $t = 5$ for \$20,000

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- The corporate tax rate is 21%, and the discount rate is 8%
- Should the company buy the copier or not?

Example, Putting It All Together

Year	0	1	2	3	4	5
Gross profit		20,000	20,000	20,000	20,000	20,000
Depreciation		16,000	16,000	16,000	16,000	16,000
EBIT		4,000	4,000	4,000	4,000	4,000
Net income		3,160	3,160	3,160	3,160	3,160
+ Depreciation		16,000	16,000	16,000	16,000	16,000
Operating cash flows		19,160	19,160	19,160	19,160	19,160
Capital investment	−100,000					
Sale of equipment						20,000
Working capital	−10,000					10,000
Total cash flows	−110,000	19,160	19,160	19,160	19,160	49,160
NPV	−13,082					

Example, Putting It All Together

Year	0	1	2	3	4	5
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Depreciation		16,000	16,000	16,000	16,000	16,000
EBIT		4,000	4,000	4,000	4,000	4,000
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- The Excel solution is in the "Copier" tab of Module 7 Excel Example.xlsx

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  - **Operating cash flow before accounting for ΔNWC**, in years 1–5 expenses fall by \$20,000, adding to gross profits. Depreciation rises by $(100{,}000 - 20{,}000)/5 = \$16{,}000$ per year, a net \$4,000 higher EBIT. After taxes and adding back depreciation, operating cash flow rises by \$19,160 each year

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- Putting it all together, NPV $= -\$13{,}082$, so the company should not buy the new machine

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- Total project cash flow = capital investments + operating cash flow before accounting for ΔNWC + changes in working capital

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- Lay the years out in a table as in the copier example, then discount the total cash flows

Module 7, Part 2

The Principles behind Project Cash Flows

General Principles for Identifying Project-Level Cash Flow

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4. Only consider incremental cash flows

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- In the copier table, we discounted the operating cash flows (19,160 a year), not the net income (3,160)

Cash Flow vs. Accounting Income

- A project costs $2,000 and is expected to last 2 years, producing cash income of $1,500 and $500, respectively. The cost can be depreciated at $1,000 per year. Given a 10% required return, compare the NPV using accounting income to the NPV using cash flow

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- This project is clearly a loser!

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- The copier table has no interest line, even if the company borrows to buy the machine

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- Different depreciation schedules, in this course we stick to "straight-line" depreciation, as in the copier, $(100{,}000 - 20{,}000)/5 = \$16{,}000$ a year

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$$\text{Incremental cash flow} = \text{Cash flow with the project} - \text{Cash flow without it}$$

- In the copier table, the labor savings and the extra paper supply exist only because of the machine, so they count

Incremental Cash Flows, a Few Rules That Are Easy to Forget

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b. Ignore sunk costs
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d. Beware of allocated overhead costs
e. Recognize investments in working capital
f. Remember terminal cash flows

Rule a, Indirect Effects

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Rule a, Indirect Effects

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- Nintendo plans a new-generation gaming console. It requires an investment of \$200m and produces cash flows of \$400m and \$800m in years 1 and 2. Looks like a great investment!





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- But the launch will hurt sales of the older console, forecast to create $800m and $1,000m in years 1 and 2, only $500m and $400m with the new one on the market





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- But the launch will hurt sales of the older console, forecast to create $800m and $1,000m in years 1 and 2, only $500m and $400m with the new one on the market
- What is the **incremental** cash flow of the new console?





Rule a, Indirect Effects

- Solution ($ millions),

	$t = 0$	$t = 1$	$t = 2$
Cash flow from the old console without the new one		800	1,000
Cash flow from the old console with the new one		500	400
Cash flow from investment in and sales of the new one	-200	400	800
Incremental cash flow of the new console	-200	$400 - 300 = 100$	$800 - 600 = 200$

Rule b, Sunk Costs

- **Sunk costs**, expenses already paid, or which cannot be changed by the investment decision
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- Many sunk costs are "related" to a project, but no longer incremental if the current decision does not affect them



Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost $0.6 million and was completed last year
- Launching requires another $1 million investment (marketing and so on), with a present value of future operating cash flows of $1.5 million

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- Is this project a good idea today?
- Was this project a good idea last year?

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- E.g., you plan a new factory building that costs \$10m to construct, on a piece of land you already own (and do not need for anything else). Alternatively, you could sell the land for \$3m
- What is the actual cost of building the factory?

Rule d, Overhead

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  - E.g., rent and utilities of the central office, top management salaries

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- The principle of incremental cash flows says only *changes* in overhead that result from a project count

Rule e, Working Capital

- If a project requires additional investment in net working capital (and most do!), those cash flows belong in the analysis
  - Easy to forget! (The copier example had them)

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  - Only changes in NWC that are incremental to the project count
- Working capital often keeps changing during the project, and is often recovered at the end
- Common mistakes,
  1. Forgetting about working capital entirely
  2. Forgetting that working capital may change during the life of the project
  3. Forgetting that working capital is often recovered at the end of the project

Rule f, Terminal Cash Flows

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- E.g., in the copier example, suppose the machine could be sold for \$50,000 at $t = 5$. It has been depreciated to \$20,000, so the company pays taxes on the gain,

$$(50{,}000 - 20{,}000) \times 21\% = \$6{,}300$$

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$$(50{,}000 - 20{,}000) \times 21\% = \$6{,}300$$

- The net (after-tax) proceeds are $50{,}000 - 6{,}300 = \$43{,}700$, and now the NPV is positive ($+\$3{,}048$)

Example, the Copier with a Higher Terminal Cash Flow

Year	0	1	2	3	4	5
Gross profit		20,000	20,000	20,000	20,000	20,000
Depreciation		16,000	16,000	16,000	16,000	16,000
EBIT		4,000	4,000	4,000	4,000	4,000
Net income		3,160	3,160	3,160	3,160	3,160
+ Depreciation		16,000	16,000	16,000	16,000	16,000
Operating cash flows		19,160	19,160	19,160	19,160	19,160
Capital investment	−100,000					
Sale of equipment (after tax)						43,700
Working capital	−10,000					10,000
Total cash flows	−110,000	19,160	19,160	19,160	19,160	72,860
NPV	3,048					
IRR	8.8%					

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- The Excel solution is in the "Copier Higher Terminal" tab of Module 7 Excel Example.xlsx

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- E.g., oil wells require "capping" when they no longer produce. Capping is expensive, so oil well projects are worth more if you do not cap them (although this is illegal!)



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- Terminal cash flows can also be negative, e.g., if we have to pay to shut down the project
- E.g., oil wells require "capping" when they no longer produce. Capping is expensive, so oil well projects are worth more if you do not cap them (although this is illegal!)
- But apparently many companies still do this, Bloomberg, "It Pays to Not Pay Your Debts"



Practice Problem

- A firm plans a new manufacturing plant on a site it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. The market value of the site
b. The market value of the existing buildings
c. Demolition costs and site clearance
d. The cost of a new access road put in last year
e. Lost cash flows on other projects due to executive time spent on the new facility
f. Future depreciation of the new plant
g. The reduction in the firm's tax bill resulting from depreciation of the new plant
h. The initial investment in inventories of raw materials
i. Money already spent on engineering design of the new plant

Practice Problem

a, b. The site and buildings could have been sold or put to another use. Their values are opportunity costs, treated as incremental cash outflows

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i. The expenditure on design is a sunk cost and not an incremental cash flow

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- Only incremental cash flows count, would the cash flow exist without the project?
- Count indirect effects, opportunity costs, working capital, and terminal cash flows (with the tax on any gain over book value), ignore sunk costs and allocated overhead

Module 7, Part 3

Full-Scale Capital Budgeting Problems

How to Attack a Full-Scale Problem

1. Build the project-level income statement year by year, then get the operating cash flow before accounting for ΔNWC

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2. Lay out the working capital each year, the *changes* in NWC are the cash flows
3. Add the capital investment at the start and the terminal cash flows at the end (after tax)
4. Total the cash flows year by year, discount at the discount rate, and apply the NPV rule

Example 1, Growing Sales and a Taxed Sale of Equipment

- A company makes bottles and just developed a new bottle design
- The initial investment in machinery is \$6 million. The equipment is depreciated straight-line over 6 years to zero, but will actually be sold in 6 years for \$620,000 (so there are tax consequences)

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- The new design is phased out after 6 years. The tax rate is 21%, and the required rate of return is 12%
- What is the project's NPV?

Example 1, Growing Sales and a Taxed Sale of Equipment

- Operating cash flows ($ thousands),

Year	1	2	3	4	5	6
Sales (thousand bottles)	1,000	1,050	1,103	1,158	1,216	1,276
Sales ($)	4,000	4,200	4,410	4,631	4,862	5,105
Cost ($)	2,000	2,100	2,205	2,315	2,431	2,553
Depreciation	1,000	1,000	1,000	1,000	1,000	1,000
EBIT	1,000	1,100	1,205	1,315	1,431	1,553
Taxes	210	231	253	276	301	326
Net income	790	869	952	1,039	1,130	1,227
+ Depreciation	1,000	1,000	1,000	1,000	1,000	1,000
Operating cash flows	1,790	1,869	1,952	2,039	2,130	2,227

Example 1, Growing Sales and a Taxed Sale of Equipment

- Terminal cash flow from the sale of the equipment ($ thousands), the book value is zero so the whole sale price is taxed,

Terminal cash flows	$
Sales price	620
Tax	130
Net income from sale	490

Example 1, Growing Sales and a Taxed Sale of Equipment

- Total cash flows ($ thousands),

Year	0	1	2	3	4	5	6
Capital	−6,000						490
Working capital	−250						250
Operating cash flows		1,790	1,869	1,952	2,039	2,130	2,227
Total cash flows	−6,250	1,790	1,869	1,952	2,039	2,130	2,966
PV	−6,250	1,598	1,490	1,389	1,296	1,209	1,503
NPV	2,235						

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- The Excel solution is in the "Bottles" tab of Module 7 Excel Example.xlsx

Example 2, Working Capital That Moves with Sales

- A company is considering a new pop-up restaurant, with projected annual sales of \$10m, \$11m, and \$12m over its three years before shutting down
- The annual costs for food, personnel, rent, and so on are 70% of revenues

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- The discount rate is 11% and the tax rate is 21%
- What is the NPV of this project?

Example 2, Working Capital That Moves with Sales

- Operating cash flows and working capital ($ millions),

Year	0	1	2	3
Revenues		10	11	12
Costs		7	7.7	8.4
Depreciation		1.5	1.5	
EBIT		1.5	1.8	3.6
Taxes		0.315	0.378	0.756
Net income		1.185	1.422	2.844
+ Depreciation		1.5	1.5	0
Operating cash flows		2.685	2.922	2.844
Working capital	1	1.1	1.2	
Change in working capital	1	0.1	0.1	−1.2

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Operating cash flows		2.685	2.922	2.844
Working capital	1	1.1	1.2	
Change in working capital	1	0.1	0.1	−1.2

- Working capital is bought at the start of the year (the end of the previous year)

Example 2, Working Capital That Moves with Sales

- Total cash flows ($ millions),

Total cash flows	0	1	2	3
Capital	−3			
Working capital	−1	−0.1	−0.1	1.2
Operating cash flows		2.685	2.922	2.844
Total cash flows	−4	2.585	2.822	4.044
PV	−4	2.329	2.290	2.957
NPV	3.576			

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- The Excel solution is in the "Pop-up Restaurant" tab of Module 7 Excel Example.xlsx

Example 3, Depreciation That Outlives the Product

- A company is considering a new wireless home networking gadget. Sales and cost forecasts,
  - Annual sales of 50,000 units per year for each of **four** years of product life
  - Wholesale price $260, unit production cost $110, ready to ship in one year

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  - New equipment for $10 million, depreciated straight-line over a **5-year** life to zero, then scrapped
  - A testing and support lab costing $2.8 million per year (personnel and so on)

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  - Accounts receivable plus inventory of 15% of each year's sales, and accounts payable of 15% of the year's cost of goods sold, both in place at the start of each year

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  - Accounts receivable plus inventory of 15% of each year's sales, and accounts payable of 15% of the year's cost of goods sold, both in place at the start of each year
- The marginal tax rate is 21%, and the discount rate is 12%
- What is the NPV of this project?

Example 3, Depreciation That Outlives the Product

- Operating cash flows ($ thousands),

Year	1	2	3	4	5
Revenues	13,000	13,000	13,000	13,000	
Costs (variable)	5,500	5,500	5,500	5,500	
Lab costs	2,800	2,800	2,800	2,800	
Depreciation	2,000	2,000	2,000	2,000	2,000
EBIT	2,700	2,700	2,700	2,700	−2,000
Taxes	567	567	567	567	−420
Net income	2,133	2,133	2,133	2,133	−1,580
+ Depreciation	2,000	2,000	2,000	2,000	2,000
Operating cash flows	4,133	4,133	4,133	4,133	420

Example 3, Depreciation That Outlives the Product

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Revenues	13,000	13,000	13,000	13,000	
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EBIT	2,700	2,700	2,700	2,700	−2,000
Taxes	567	567	567	567	−420
Net income	2,133	2,133	2,133	2,133	−1,580
+ Depreciation	2,000	2,000	2,000	2,000	2,000
Operating cash flows	4,133	4,133	4,133	4,133	420

- Depreciation continues in year 5, a tax shield in that year, so the forecast includes year 5

Example 3, Depreciation That Outlives the Product

- Working capital ($ thousands), in place at the start of each year,

Year	0	1	2	3	4
Accounts receivable and inventory	1,950	1,950	1,950	1,950	0
Accounts payable	825	825	825	825	0
NWC	1,125	1,125	1,125	1,125	0
Change in NWC	1,125	0	0	0	−1,125
Cash flow effect	−1,125	0	0	0	1,125

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Cash flow effect	−1,125	0	0	0	1,125

- Accounts receivable plus inventory, $15\% \times 13{,}000 = 1{,}950$, accounts payable, $15\% \times 5{,}500 = 825$

Example 3, Depreciation That Outlives the Product

- Total cash flows ($ thousands),

Year	0	1	2	3	4	5
Equipment	−10,000					
Working capital	−1,125	0	0	0	1,125	
Operating cash flows		4,133	4,133	4,133	4,133	420
Total cash flows	−11,125	4,133	4,133	4,133	5,258	420
PV of cash flows	−11,125	3,690	3,295	2,942	3,342	238
NPV	2,382					
IRR	21%					

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- The Excel solution is in the "Gadget" tab of Module 7 Excel Example.xlsx

Example 3, Extension 1, More Efficient Working Capital

- What happens to the NPV if the firm uses working capital more efficiently, accounts receivable plus inventory of 5% of sales and accounts payable of 5% of the cost of goods sold? For example,
  - Finished goods used to sit in inventory for 3 months before selling, with an upgraded inventory system they sell 1 month after production, inventories are now 1/3 of before
  - Payment used to be collected 3 months after a sale on average, with a more capable collection officer bills are collected in 1 month, accounts receivable are now 1/3 of before

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- The NPV increases to $2,655 thousand
- More efficient use of NWC increases firm value! (Excel solution in the "Gadget Efficient NWC" tab)

Example 3, Extension 2, Bonus Depreciation

- **Bonus depreciation**, a tax rule that lets businesses deduct the cost of qualifying assets immediately instead of over time



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- Would the NPV increase or decrease?

- The NPV increases to $2,968 thousand
- A bonus for the firm because the tax savings from depreciation are front-loaded! (Excel solution in the "Gadget Bonus Depreciation" tab)


ARTICLE

100% bonus depreciation returns with the One, Big, Beautiful Bill

Authors: Jeremy Sompels, Jonathan Winterkorn

July 21, 2025 / 4 min read

Example 3, Extension 2, Without Bonus Depreciation

- Operating cash flows without bonus depreciation ($ thousands, years 2 and 3 repeat year 1),

Year	0	1	···	4	5
Revenues		13,000	···	13,000	
Costs (variable)		5,500	···	5,500	
Lab costs		2,800	···	2,800	
Depreciation		2,000	···	2,000	2,000
EBIT		2,700	···	2,700	−2,000
Taxes		567	···	567	−420
Net income		2,133	···	2,133	−1,580
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Net income		2,133	···	2,133	−1,580
+ Depreciation		2,000	···	2,000	2,000
Operating cash flows		4,133	···	4,133	420

- The depreciation tax shield is $2{,}000 \times 21\% = 420$ in each of years 1–5

Example 3, Extension 2, With Bonus Depreciation

- Operating cash flows with bonus depreciation ($ thousands, years 2 and 3 repeat year 1),

Year	0	1	⋯	4	5
Revenues	0	13,000	⋯	13,000	
Costs (variable)	0	5,500	⋯	5,500	
Lab costs	0	2,800	⋯	2,800	
Depreciation	10,000	0	⋯	0	0
EBIT	−10,000	4,700	⋯	4,700	0
Taxes	−2,100	987	⋯	987	0
Net income	−7,900	3,713	⋯	3,713	0
+ Depreciation	10,000	0	⋯	0	0
Operating cash flows	2,100	3,713	⋯	3,713	0

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Depreciation	10,000	0	⋯	0	0
EBIT	−10,000	4,700	⋯	4,700	0
Taxes	−2,100	987	⋯	987	0
Net income	−7,900	3,713	⋯	3,713	0
+ Depreciation	10,000	0	⋯	0	0
Operating cash flows	2,100	3,713	⋯	3,713	0

- The whole tax shield, $10{,}000 \times 21\% = 2{,}100$, arrives at $t = 0$, front-loaded

Practice Problem

- A firm has developed a new mousetrap. It can go into production for an initial investment in equipment of \$6.3 million. The equipment will be depreciated straight-line over 6 years, but in fact can be sold after 6 years for \$695,000
- Working capital at each date must be maintained at 10% of next year's forecast sales. Production costs are \$1.80 per trap and the traps sell for \$7 each. The project ends in 6 years when the trap becomes obsolete. The tax rate is 40%, and the required rate of return is 11%

Year	0	1	2	3	4	5	6	After
Sales (million traps)	0	0.6	0.8	0.9	0.9	0.5	0.2	0

Practice Problem

- Suppose the firm can cut its working capital requirements in half with better inventory control. By how much does this increase the project NPV?

Practice Problem

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- Solution ($ millions), working capital at each date is 10% of next year's sales,

Year	0	1	2	3	4	5	6
Sales units		0.600	0.800	0.900	0.900	0.500	0.200
Revenue	0.000	4.200	5.600	6.300	6.300	3.500	1.400
NWC	0.420	0.560	0.630	0.630	0.350	0.140	0.000
Cash flow from NWC	−0.420	−0.140	−0.070	0.000	0.280	0.210	0.140

Practice Problem

- With working capital requirements halved ($ millions),

Year	0	1	2	3	4	5	6
New NWC	0.210	0.280	0.315	0.315	0.175	0.070	0.000
New cash flow from NWC	−0.210	−0.070	−0.035	0.000	0.140	0.105	0.070
Change in cash flow	0.210	0.070	0.035	0.000	−0.140	−0.105	−0.070

Practice Problem

- With working capital requirements halved ($ millions),

Year	0	1	2	3	4	5	6
New NWC	0.210	0.280	0.315	0.315	0.175	0.070	0.000
New cash flow from NWC	−0.210	−0.070	−0.035	0.000	0.140	0.105	0.070
Change in cash flow	0.210	0.070	0.035	0.000	−0.140	−0.105	−0.070

- Discounting the changes in cash flow at 11%, the change in NPV is $0.1095 million

Practice Problem

- A firm may upgrade its modem pool. It last upgraded 2 years ago, when it spent $140 million on equipment with an assumed life of 5 years and an assumed salvage value of $20 million for tax purposes. The firm uses straight-line depreciation. The old equipment can be sold today for $100 million
- A new modem pool can be set up today for $210 million. It has a 3-year life and is depreciated to zero straight-line. The new equipment increases sales by $28 million per year and decreases operating costs by $14 million per year. At the end of 3 years it is worthless. The tax rate is 30% and the discount rate is 9%

Practice Problem

a. What is the net cash flow at time 0 if the old equipment is replaced?
b. What are the incremental cash flows in years 1, 2, and 3?
c. What is the NPV of the replacement project?
d. What is the IRR of the replacement project?

Practice Problem

- a. Annual depreciation is $(\$140 - \$20)/5 = \$24$. Book value at the time of sale is $\$140 - 2 \times \$24 = \$92$

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- a. Annual depreciation is $(\$140 - \$20)/5 = \$24$. Book value at the time of sale is $\$140 - 2 \times \$24 = \$92$

Sales price = $100, so net-of-tax proceeds from the sale are:
$100 − [0.30 × ($100 − $92)] = $97.60

Therefore, the net cash outlay at time 0 is $210 − $97.60 = −$112.40.

b. The project saves $14 in operating costs and increases sales by $28. The depreciation expense for the new machine would be $70 per year.

Therefore, including the depreciation tax shield, operating cash flow increases by:
($28 + $14) × (1 − 0.30) + ($70 × 0.30) = $50.40 per year

c. NPV = −$112.40 + $50.40 × ((1 ÷ 0.09) − {1 ÷ [0.09(1 + 0.09)^3]})
= $15.18

d. To find the internal rate of return, find the interest rate "r" that makes the Present Value of the inflows equal and opposite to the initial outflow. To do this set the PV of the three year annuity to $112.40 and solve for the discount rate (r):

$112.40 = $50.40 × [1 ÷ r − 1 ÷ r × (1 + r)^3]
r = 16.43%

Key Takeaways

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- Use cash flows, not accounting profits, treat financing separately, and remember the depreciation tax shield
- Only incremental cash flows count, include indirect effects, opportunity costs, working capital, and terminal cash flows, ignore sunk costs
- Lay the years out in a table, total the cash flows, discount, and apply the NPV rule

Thank you!

Please let me know if you have any questions.